

Innovation on Diagnosis and Maintenance of AI Datacenters

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Agenda

- Datacenter Diagnostic & Debugging Case Study
- Major Challenges in Datacenter Debugging
- Datacenter-to-Lab Transformation
- Optimizing Issue Duplication
- Diagnostic Strategies
- System Health and Performance Monitoring
- Fault Identification and Prevention
- Expected Features on Future PCIe Device
- Future PCIe Device Architecture

Datacenter Diagnostic & Debugging Case Study

Context

- **Deployment Scale:** 200+ NVIDIA GPU Server Systems
- **Validation:** Systems passed 100% of lab validation:
  - Power & Signal Integrity (PI/SI)
  - Functional & Stress Testing (Functional/Thermal Testing)
- **Issue:** Intermittent system hangs in datacenter

The Datacenter Mystery

BMC logs showed zero anomalies—no thermal throttling, no power drops, and no hardware alerts

Datacenter Diagnostic & Debugging Case Study

The Environment:

- Customer environments are "Living Ecosystems"
- Dynamic Resource Scheduling: Hundreds of users sharing NVIDIA GPU/CPU pools
- Software Stack Complexity: Interaction between OS, drivers, and specific user applications

Simulation Attempts:

- Identical Ubuntu OS and software stack
- Simulated multi-user environment
- Exhaustive Stressing: Concurrent memory, network, storage, and thermal “stressed tests“

Result:

- System remained stable in the lab despite these stresses

Datacenter Diagnostic & Debugging Case Study

The Discovery:

Success was finally achieved by coupling the customer’s specific software with targeted CPU stress.

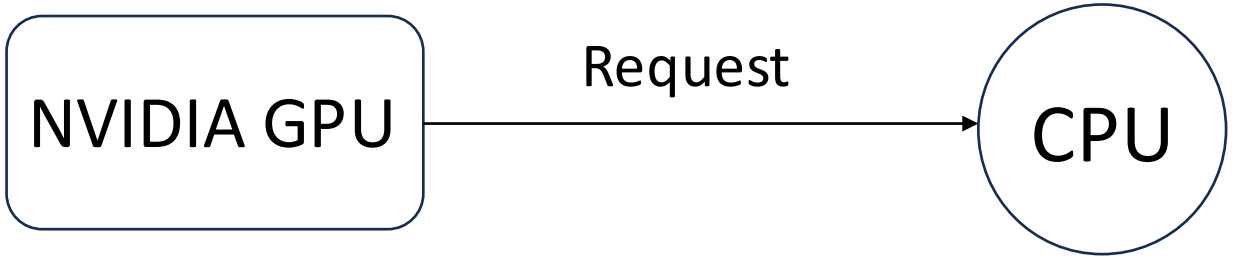
Key Insight:

The failure wasn't just about load; it was about the specific interaction between application-level communication and CPU-intensive scheduling.

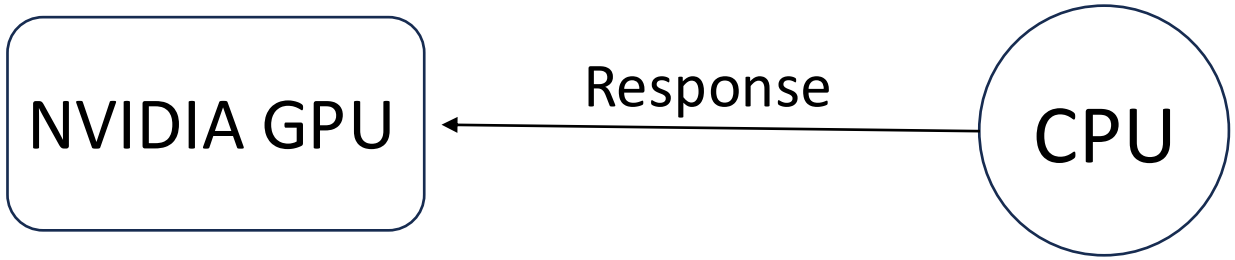
Failure Duplication

(Datacenter diagnostic & debugging case study)

- The device (GPU) send a request to the host (CPU).



- The host (CPU) takes a significant amount of time to respond.

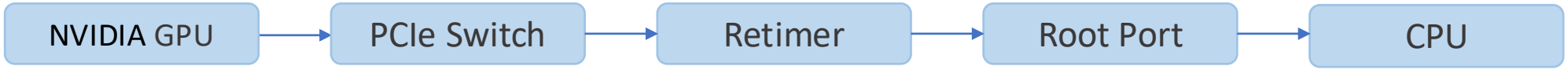


- PCIe port completion timeout

PCIe Port Completion Timeout Analysis

(Datacenter diagnostic & debugging case study)

- The Critical Path:



- The Trigger: Systemic latency spikes during peak compute loads.
- The Crash: Host response lag exceeds the threshold, triggering a PCIe Completion Timeout (CTO) and a total system hang.
- The Configuration Gap:
  - Root Ports: Configurable via UEFI BIOS, allowing for increased timeout tolerance.
  - Endpoints: Effectively "Black Boxes." Timeouts are coded in vendor firmware, leaving motherboard designers without visibility or control.

Major Challenges in Datacenter Debugging

- Limited Access and Time
- Ambiguous Symptom
- Duplication Difficulty
- Physical Measurement Limitations
- Prohibitive Cross-Swapping
- Infeasible Trial-and-Error
- Status Registers Access Restriction

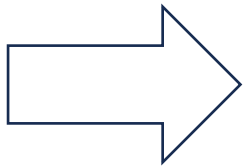
Datacenter-to-Lab Transformation

- Trace Dump
- Crash Dump
- Register Dump
- State Dump
- System Logs
- Firmware Logs
- Application Logs

- Cross-Layer Timestamp Alignment
- Inter Node Cluster Synchronization
- Sideband and In-Band Event Merging
- Telemetry Data Stream Concurrency

- Protocol-Level State Tracking
- Signal Integrity Drop Mapping
- Pattern-Based Signature Matching
- Pre-Failure Workload Characterization

Lab Replication



Comprehensive Log Collection

Multi-Dimensional Time Correlation

Root-Cause Trigger Analysis

Optimizing Issue Duplication

Minimizing Configuration

- Reduce Components of PCIe Cards, Hard Drives, and Switch Boards
- Reduce the Quantity of CPUs, Memory Modules, and GPUs

Maximizing Failure Frequency

- Stress CPU
- Stress Memory
- Stress Network
- Stress Storage
- Stress Thermal

Perspectives: 4 Common Datacenter Failures

Hardware Failure



Examples: Power supply unit (PSU) burnout, disk drive crashes, memory (RAM) defects, or cooling system malfunctions.

Firmware Failure



Examples: BIOS corruption, outdated microcode, or incompatible RAID controller logic.

Driver Failure



Examples: Network Interface Card (NIC) driver crashes, kernel panics caused by incompatible updates, or memory leaks in I/O drives.

Application Failure



Examples: Database deadlocks, memory overflows in virtualization software, or logic errors in orchestration tools.

Three Stages of Datacenter GPU Server Failure

Pre-Boot

Power rails begin ramping up, clocks start oscillating, and components initialize.

Booting

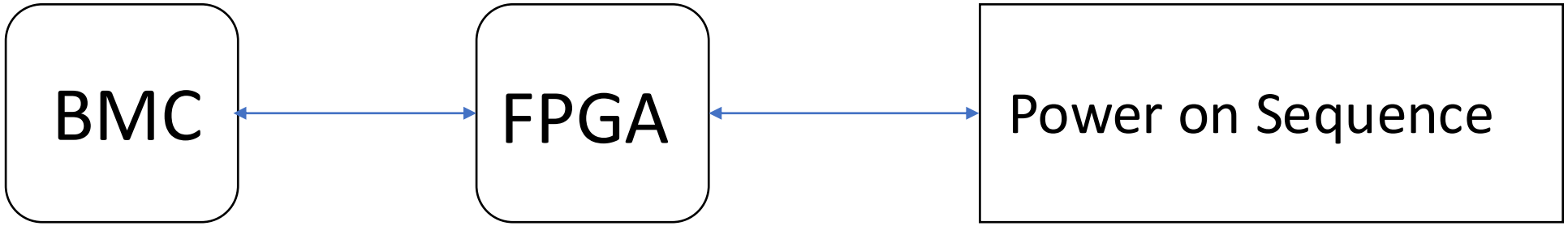
The CPU begins to execute BIOS codes.

Post-Boot

The operating system is loaded and running.

Diagnosis at Pre-Boot Stage

- FPGA should have full control over the power on sequence.
- FPGA should latch the power status of the individual power.
- FPGA should detect the status of clock and reset signals.
- BMC can access to the FPGA’s registers.



Diagnosis During Booting

- BIOS of the motherboard will send the progress code to BMC. BMC will display the progress code on web interface.
  - Send boot progress code (NetFn 2Ch, Command 02h)
  - Get boot progress code (NetFn 2Ch, Command 03h

```
root@gb200nvl-bmc-avi:~# ipmitool raw 0x2C 0x03 0xAE
 ae 01 00 00 00 19 10 10 03 00
root@gb200nvl-bmc-avi:~#
```

- BIOS of the motherboard will print the debugging information to the serial port. BMC will redirect the serial port to networking.

Diagnosis after Boot Up

PCIe error reporting

Two Error Reporting Paradigms

- (1) Baseline Capability
- (2) Advanced Error Reporting Capability

Error Classification

- (1) Correctable errors
- (2) Uncorrectable errors
  - Non-Fatal
  - Fatal

Correctable Error Category

- Receiver error
- Bad TLP
- Bad DLLP
- Replay_Numrollover
- Replay timer timeout
- Advisory non-fatal error
- Corrected internal error
- Header log overflow

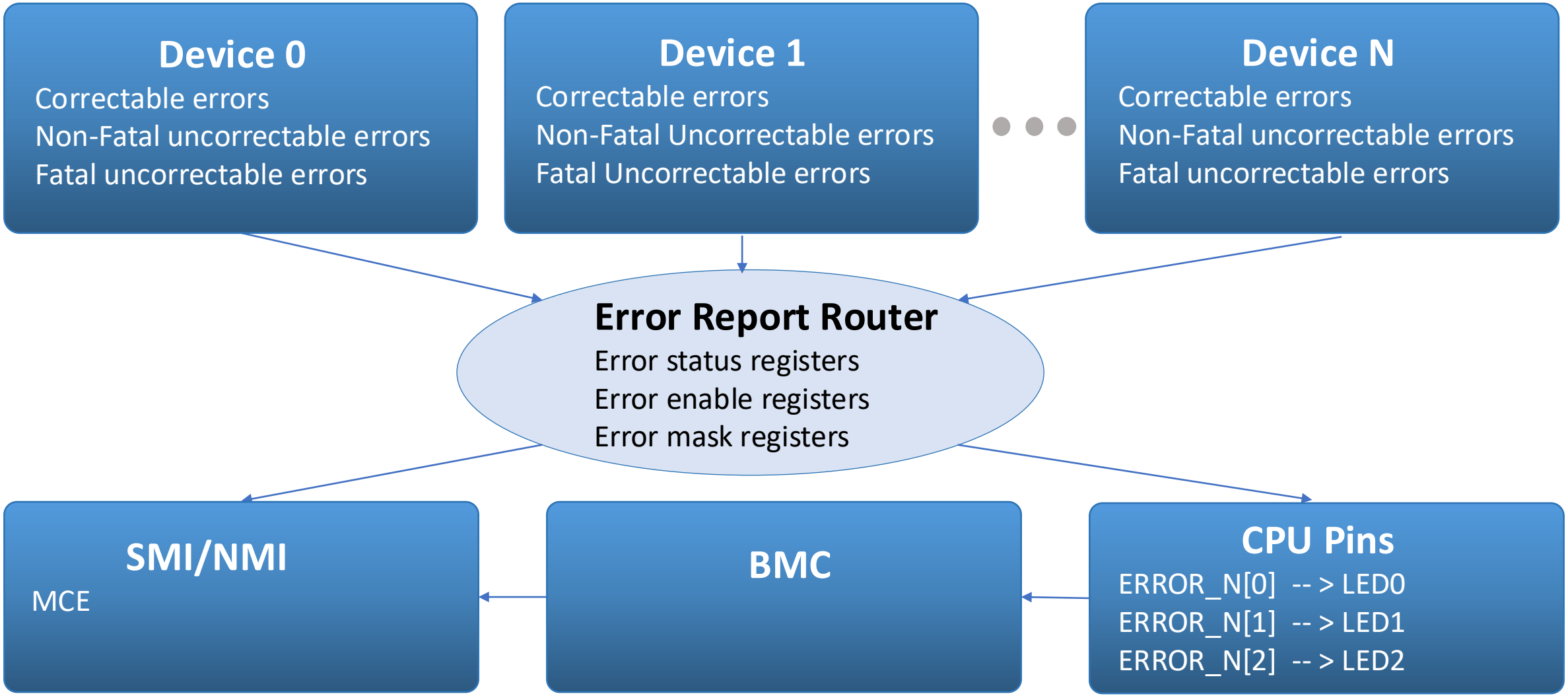
Non-Fatal Uncorrectable Errors

- TLP Prefix Blocked Error
- AtomicOp Egress Blocked Status
- MC Blocked TLP Status
- ACS Violation Status
- Unsupported Request Error
- ECRC Error Status
- Unexpected Completion Status
- Completer Abort Status
- Completion Timeout Status
- Poisoned TLP Status

Fatal Uncorrectable Errors

- Uncorrectable Internal Error
- Malformed TLP Status
- Receiver Overflow Status
- Flow Control Protocol Error Status
- Surprise Down Error Status
- Data Link Protocol Error

Error Report Route



System Health and Performance Monitoring

Hardware Monitoring

- PCIe configuration and status register tracking.
- Voltage, current, and power consumption measurement.
- Temperature monitoring for CPUs, GPUs and peripherals.

Cooling System Management

- Fan speed control and monitoring.
- Liquid pressure, flow rate, and in/out temperatures reporting.
- Leakage detection.

Performance Evaluation

- Performance monitoring for Memory, Storage, CPU, and GPU.

Data Processing

Comprehensive Hardware Metrics Extraction

- Real-Time Status & Configuration Register Dumps
- Continuous Hardware Performance Data Recording
- Environmental Monitoring Data Stream Logging

Device Failure Risk Modeling

- Dynamic Weight Allocation for Multivariable Hardware Inputs
- Custom Cost Functions Specialized for Diverse Component Types
- Adaptive Threshold Tuning for Complex System Topologies

Model Training Using Historical Data

- Supervised Learning on Legacy Failure Signature Datasets
- Continuous Training Utilizing Real-World Measurement Data
- Cross-Validation Across Heterogeneous Hardware Clusters

Itemized Device Failure Prediction

- Per-Component Risk Scoring and Degradation Probability
- Proactive Migration and Early-Warning Triggers
- Automated Remediation Triggers via Out-of-Band Actions

Preemptive Action Strategies

Compute Fleet Workload Evacuation

- Live Virtual Machine Migration to Healthy Target Hosts
- Proactive Stateless Task Re-routing

Memory Modules (DIMMs)

- Early Isolation to Prevent System Hung
- Scheduled Replacement

Storage Infrastructure (HDD/SSD)

- Pre-Failure Data Mirroring
- Scheduled Swapping of High-Risk Storage Drives

Power Supply Units (PSU)

- Load-Balancing Adjustments
- Proactive Unit Swapping

Liquid Cooling Systems

- Automated Flow-Rate Compensation
- Scheduled Manifold Cleaning

Diagnosis Dilemma

Communication Loss

- Missing GPUs render internal registers inaccessible.
- System hangs prevent dumping configuration and status registers.

Power & Reset Events

- System reset could clear the configuration and status register settings.
- Power failures destroy the debugging context.

Lack of Traceability

- Failing to capture transient processes hinders root cause tracing.

Expected Features on Future PCIe Device

Breaking the "Black Box"

- Enable BMC access to vital PCIe registers via the I2C interface.
- Standardize register window set of black box across specific PCIe device classes.

Secure Debugging Context

- Preserve debugging context during system hangs or main power loss.

Capturing the Failure Process

- Record the change history of the PCIe configuration and status registers.

Future PCIe Device Architecture

Dual-Path Connectivity

- In-Band Path access to the PCIe configuration and status registers.
- Out-of-Band (OOB) Path alternative access via BMC interface .

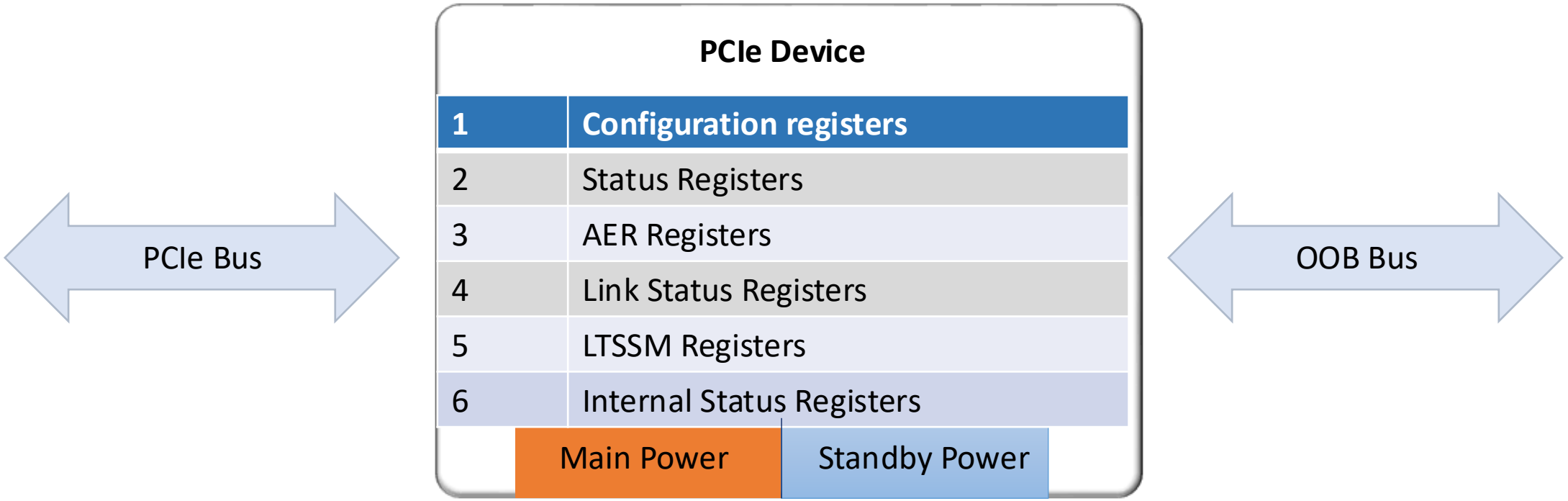
Standby-Powered Register Domain

- Register status retention even when the main power is down.

Register Change History Visibility

- Preservation of the chronological change history of these registers.

Future PCIe Device Diagram



Conclusion: Redefining RAS in the AI Era

From Reactive to Predictive

- Use monitoring data to predict device failure.
- Maintain the system based on the prediction before it fails.

The Path Forward: Out-of-Band (OOB) Standard

- Open a window on the device end for debugging.
- Standardize the visibility to break "Black Box".

Call to Action

- Expand the PCIe specification to support OOB access to configuration and status registers.

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